

Driver Safety AI - Academic Session Report

Real-Time Driver Drowsiness and Vigilance Detection System

Project Owner:	Manoj Kumar C (B.Tech CSE)
Session ID:	web_1787324808
Session Duration:	279.8 seconds
Total Frames Processed:	29
Average System FPS:	21.4 FPS
Minimum System FPS:	6.7 FPS

Deep Learning Architecture & Feature Configuration

Model Architecture:	1D CNN (32 filters, kernel 3) + 2-layer Bi-LSTM (64 hidden, bidirectional) + Dense Classifier
Temporal Input Tensor:	30 frames × 12 features (Shape: Batch × 30 × 12)
Features Extracted:	EAR_LEFT, EAR_RIGHT, MEAN_EAR, MAR, YAW, PITCH, ROLL, PERCLOS, BLINK_RATE, EYE_CLOSURE_DUR, MOUTH_OPEN_DUR, HEAD_MOTION_MAG
Output Target Classes:	ALERT, DROWSY, YAWNING, DISTRACTED

Session Telemetry & Risk Index Statistics

Metric / Parameter	Value / Statistic
Average DVI Risk Score:	7.1%
Maximum Peak DVI Risk Score:	7.7%
Voice Warning Events:	0
Audio Alarm Events:	0
Recovery Events:	0
Total Recovery Time:	0.0 seconds
Average Recovery Time:	0.0 seconds
Total Blinks Detected:	0
Total Yawns Detected:	0
Frames in ALERT state:	29
Frames in DROWSY state:	0
Frames in YAWNING state:	0
Frames in DISTRACTED state:	0

Academic Disclaimer: This session report was automatically generated by Driver Safety AI. The Driver Vigilance Index (DVI) is a project-defined academic score designed for experimental evaluation and is not an automotive-certified safety metric.